



Pain Description Matters!

The Effect of Identity and Language Choice on Physicians' Perceptions of Pain

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Contact Me!



Key Question

How do physicians assess pain based on a patient's race, gender, and language choice?

What's the connection?

Physicians have **negative** perceptions of a patient (PT) based on race and gender

Negative Perceptions of PTs

Black (vs. White) PTs

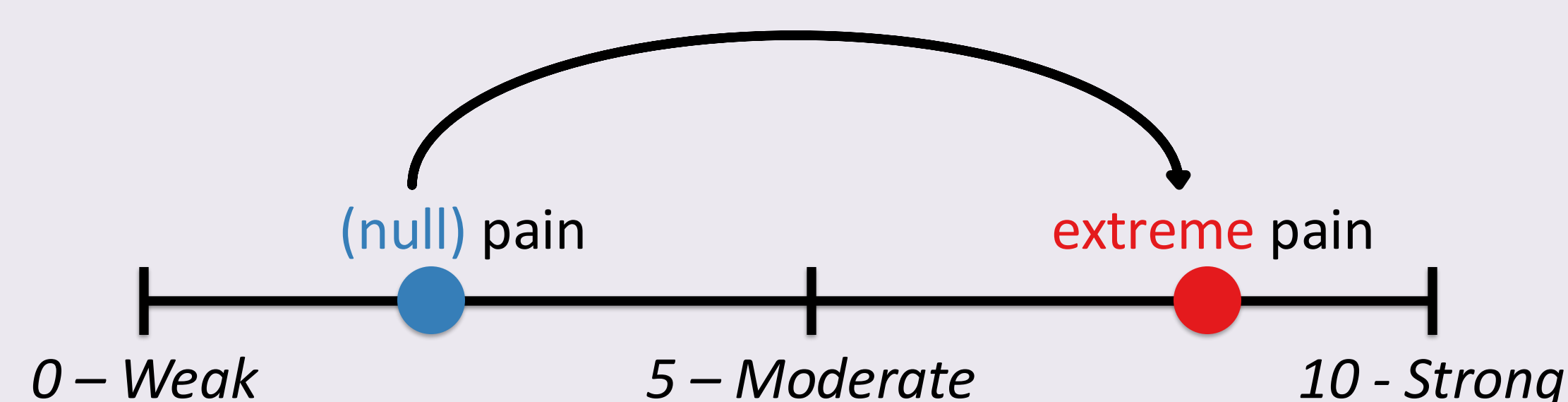
- Overly high pain tolerance¹
- Understate pain¹
- Distrusted pain reports²

Women (vs. Men) PTs

- Low pain tolerance^{1,3}
- Exaggerate pain^{1,3}
- Distrusted pain reports³

Women and Black PTs may **emphasize** symptoms to be heard/taken seriously⁴

Intensifiers, like '**extreme**', boost the meaning of a word or phrase^{5,6}



Linguistic bias is the **negative** evaluation of speakers based on their voice, dialect, or language choices, and can lead to discrimination⁷



Speaking in a different dialect
→ less housing callbacks⁷

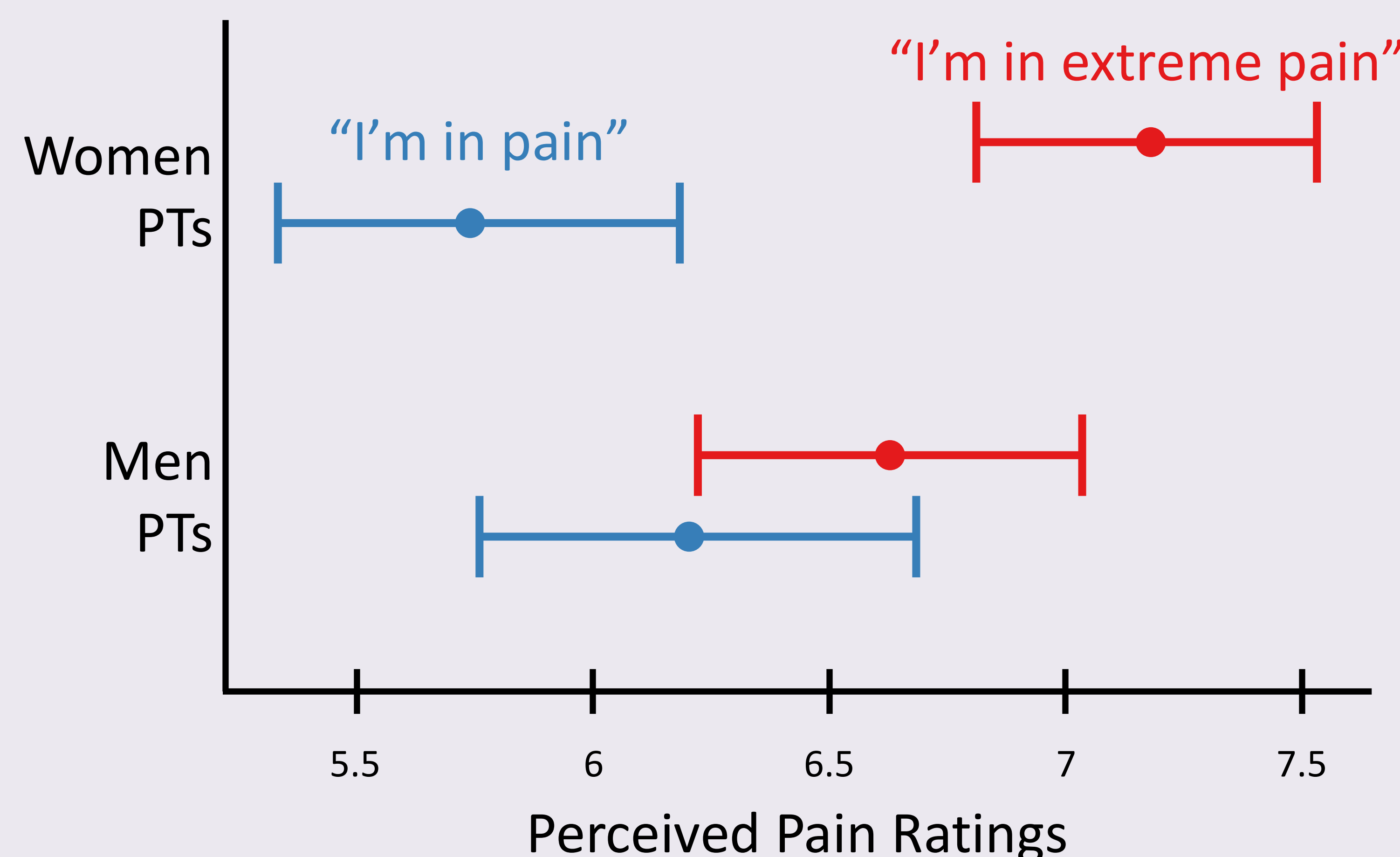


Speaking with an accent
→ less likely to be hired^{8,9}



Speaking with an accent
→ less trustworthy and credible^{10,11}

So who's in extreme pain?



Physicians rated pain differently based on language choice and gender

Intense vs. **Bare** rated **significantly different for women** but not men ($p < .01$)

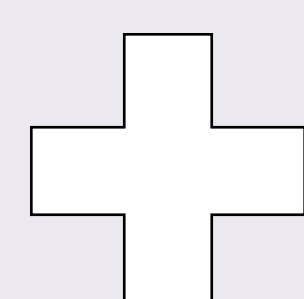
What's it all mean...

We hypothesize that physicians generally disbelieve and dismiss women for their pain¹² and potentially took women more seriously **only** when they emphasized pain

Women: "I'm in pain" 5.74/10

Women: "I'm in extreme pain" 7.19/10

Social Information (Gender)



Linguistic Choices (Intensifier Usage)

Pain perception increases +

Pain perception increases +++

Linguistic bias is another factor in assessing biased medical outcomes

A physician's perceptions of pain depending on a patient's linguistic choices could also impact health outcomes. Future work can build upon these results to develop interventions addressing linguistic bias in both undergraduate and graduate medical education programs.

How'd we do it?

Read this example:

A 56-year-old **Black woman** comes into your office. When you ask, "What brings you in today?", the patient responds with:

I've been having **extreme** pain in my lower back and down both of my legs for several months. My left side hurts more, and I also sometimes have **severe** tingling and numbness in my thighs and legs, which makes it hard to walk comfortably. It takes **intense** effort to stand and lift items. It's been limiting my activity, but I have noticed it's better when I sit and bend forward or when I use heat pads.

How much pain do you believe the patient is in?



54 physicians read a hypothetical patient's symptom report and rated perceived pain

Reports manipulated: Race (Black vs. White), Gender (man vs. woman), Intensity (**bare** vs. **intense**, e.g., "I'm in pain" vs. "I'm in **extreme** pain")

Citations & Acknowledgements



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